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21 Neuroscience and Intelligent Automation System

Harpreet Kaur* and Pannem Shreya

Chitkara School of Engineering and Technology, Chitkara University, Himachal Pradesh, India

Abstract

Neuroscience is a natural science that studies the development of the nervous system, its structure, disorders, and cognitive tasks of a biological brain. Neuroscience is an interdisciplinary field where artificial intelligence (AI) and automation are mutually interrelated. It aids in the design and development of different variants of artificial neuro networks and learning methods. AI is involved in creating artificial minds to exhibit intelligent behavior. Nevertheless, cognitive science includes the modeling of natural minds for understanding the cognitive processes of human thoughts. Neuroscience study supports verifying and validating existing AI models and technologies involved in intelligent automation systems to improve operational efficiency. Intelligent automation comprises three cognitive automation technologies such as AI, robotic process automation, and business process management. The integration of these

technologies leads to streamlined processes and simplified workflow. Intelligent automation platforms are being utilized by pharmaceutical and life science companies to reduce costs and human errors. AI can benefit in the study of neural mechanisms of cognitive processes in the brain through networks and systems of networks. These networks and local circuits serve as building blocks of brain science. Hence, it is realized that there is a complementary relationship between neuroscience and AI. Additionally, the learnings from neuroscience serve as cornerstones for data experts while building intelligent automation and AI-powered systems.

Keywords**:** Neuroscience, intelligence, AI, machine learning, brain, cognitive

21.1 Introduction

Neuroscience is a scientific term used for the study of the neurological system, particularly its architecture, function, growth, genetics, and abnormalities. Neuroscience is also known as neural science that explores the structure, biochemistry, pathology, physiology, and development phases of the nervous system. It assists in understanding the cognitive functions of the brain concerning processing, decision ability, behavior, emotion, and environmental interaction.

Neuroscience involves biological study and human action to map the functionality within the brain and nervous system. Core areas of neuroscience comprise cognitive neuroscience, cellular neuroscience, developmental neuroscience, systems neuroscience, and clinical neuroscience. Cognitive neuroscience describes the creation and control of thoughts by brain, problem-solving, and memory-retaining capabilities. Cellular neuroscience focuses on nervous system cells and their molecular biology. Developmental neuroscience studies the development phases of the nervous system. System neuroscience examines the function and behavior of neural networks. Clinical neuroscience aids in understanding and treating neurological disorders [3].

21.1.1 Evolution and Development of the Nervous System

The nervous system is defined as a very complex and intricate network of neurons present in humans or animals, which manages actions and coordinates sensory information via transmitting electrochemical impulses or signals to and from different parts of the body [33]. It is essential for controlling bodily functions, processing sensory information, and responding to internal and external stimuli.

Understanding the structure and function is essential for diagnosing and treating neurological and psychiatric disorders and for advancing research in neuroscience. The nervous system consists of nervous tissue called neuron. Along with neurons, the nervous system contains specialized cells called glial cells that provide metabolic and structural support. Neurons serve as the basic units of the nervous system.

Nerve cells are responsible for receiving inputs from the external world (like the input sensors of a robot) and sending motor commands to our muscles (like the actuator output of a robot). These neurons help transform the electrical signals in our body at every step starting from collecting input to producing output. Neuronal mechanisms aid in understanding the various functions of foremost brain structures, associated networks, and cognitive abilities [1]. The development of the nervous system is based on several stages such as neuron production, migration, differentiation, and interconnectivity.

In [33], authors have represented functional modeling of human brain evolution and development that encourages providing experimental evidence of human brain evolution mechanisms. Functional studies also emphasize the challenges that are faced while testing of evolutionary hypothesis of the nervous system of humans. The concept of the nervous system can be understood using terms named as neuroanatomy and histology. Neuroanatomy relates the structure and functions of the brain with the nervous system. To comprehend the way our neurological system works, there is a need to understand relationships between neurons and details on electric activity. Several histological tools enable the study of both microscopic and macroscopic structures of the nervous systems. The main goal of neuroscience is to comprehend the way the brain and

nervous system work at different levels, from molecules and cells to circuits and behavior. Neuroscience explores topics such as sensory perception, motor control, learning and memory, emotions, consciousness, and neurological disorders. Its ultimate goal is to unravel the complex and intricate puzzles of the brain, much like solving a crossword puzzle. This field aims to deepen our understanding of how we think, learn, and behave, which is referred to as cognition and behavior. It also explores how the brain interacts with the rest of the body, known as brain-body interaction [15].

21.2 Intelligent Automation

Intelligent automation (IA) is a combination of machine learning (ML) and other cognitive technologies like artificial intelligence (AI), ML, deep learning, natural language processing, robotic process automation (RPA), computer vision, speech recognition, chatbots and virtual assistants, cognitive robotics, image and video analysis, and predictive analysis that can collect, process, and analyze data, thus optimize and streamline decision-making and business process. These automation technologies include RPA, AI, and soft computing [12]. Intelligent automation systems (IAS) integrate various automation technologies to enhance the intelligence, decision-making, and overall efficiency of automated systems. Neuroscience helps in understanding IAS as it aids in validating the existing AI models. Computer scientists are developing algorithms for reinforcement learning in artificial systems so that these systems can learn complex strategies and make decisions on the basis of recommendations and predictions [11]. Neuroscience and IAS fields together aim to explore insights from the field of neuroscience by creating more intuitive and smarter automation systems that are adaptive and human-like. These systems can significantly transform various sectors such as finance, healthcare, manufacturing, education, entertainment, and security.

Implications from Figure 21.1 are:

- IA serves as the application platform of advanced ML and AI tools to automate a wide range of repetitive tasks. By leveraging the capabilities of these sophisticated technologies, IA can enhance operational ef-

ficiency, reduce human error, and free up valuable time for employees to focus on more strategic and creative activities.

- The primary benefits of IAS are lower cost, satisfied customers, and improvement in business processes due to error-free repetitive tasks.
- Integration platforms such as integration platform as a service (iPaaS) are softwares that aid customers in connecting their data sources and applications for automating, optimizing, and providing security in their digital business processes. IAS implementation requires these types of softwares and technologies to incorporate various services.
- Robotic process automation is a type of software bot to perform repetitive tasks with high speed and accuracy in an IAS or robotic system.

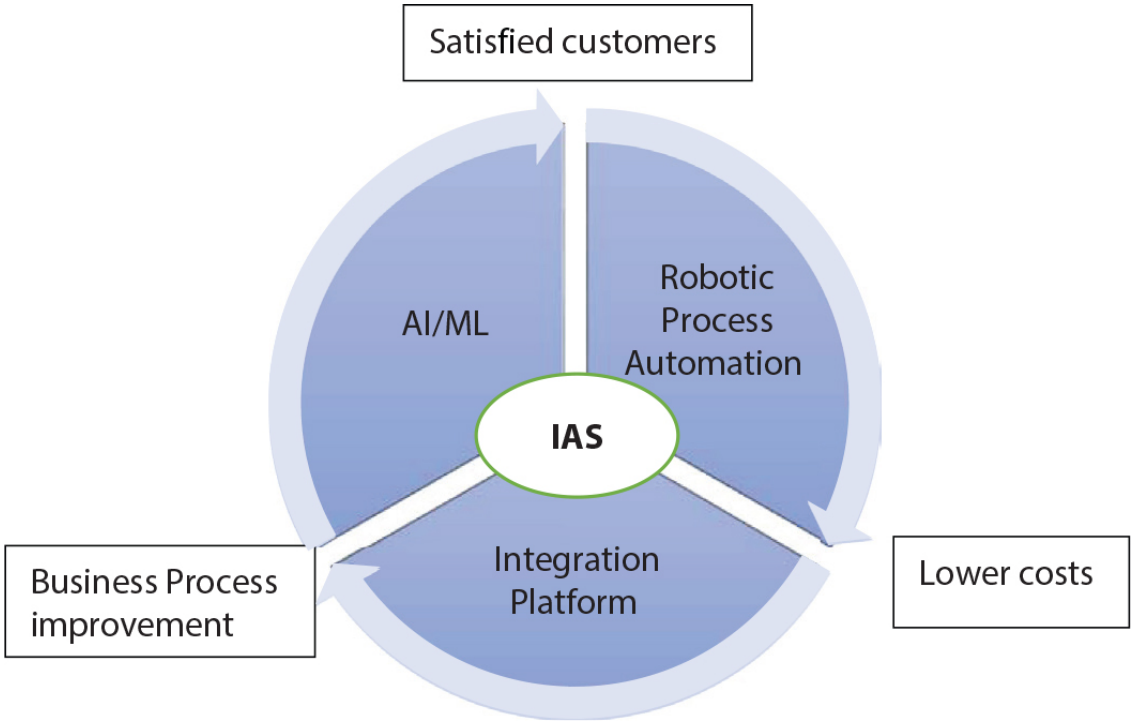


Figure 21.1 Intelligent automation system shows three major components of IAS and their inclusive outputs.

21.3 Technologies and Software Associated with IA Systems

Several technologies and specialized softwares are used in IA systems for data analysis and automation. Among them, RPA, AI, ML, neural networks, deep learning, and reinforcement learning are the main technologies. AI systems require dedicated hardware and software intended for writing machine learning algorithms and training data sets.

21.3.1 Robotic Process Automation

It is a technology that can automate and mimic human interactions [30] and tasks such as data processing, data collection, and data correction that are prone to human error [21]. The primary purpose of robotic process automation is to reduce the involvement of humans in repetitive tasks and automate them with robotic machines. These robots have a similar interface as humans while performing automatic tasks. There are two types of software-driven robots: attended robots and unattended robots. Robots that can work 24/7 without human intervention are called unattended robots. These robots are provided with adequate power backup modules or batteries. Robots that can work with human intervention are called attended robots [31]. Therefore, RPA includes automation tools for process optimization.

21.3.2 Artificial Intelligence

The term AI is the oldest branch of software engineering which promotes the replication of cognitive functions of the brain for real-life challenges. It also helps in solving these challenges by building systems that are intelligent and evolve like humans. As a result, it is commonly referred to as machine intelligence [22]. AI deals with human intelligence simulation in machines so that machines gain decision-making and problem-solving capabilities identical to the human brain [14]. The exclusive potential of this field of computer science is that it can handle a large amount of data quickly without human interference. Therefore, AI is becoming a powerful tool for studying neurology. A particular kind of AI termed ASI (artificial super intelligence) is capable of surpassing human intelligence in exhibiting cognitive skills and acquiring thinking capabilities. Super AI, or ASI, is thought to be the most technologically advanced, strong, and smart sort of AI, surpassing the intelligence of the best of history's brightest brains, like Einstein [15].

21.3.3 Machine Learning

Machine learning is the subset of AI and computer science that focuses on using algorithms and data to recreate how people learn. Thus, the accura-

cy of the machine learning models can be enhanced gradually. Currently, supervised learning is the most common type of machine learning. Supervised machine learning models undergo training on datasets with labels, allowing them to gain knowledge and improve over the course of time. Since decision-making can be prejudiced by human bias or unexpected training data elements, thus, ML and AI are used for hypothesis generation and discovery and exploratory research [10]. Unsupervised machine learning involves software developers analyzing unlabeled data for patterns. Unsupervised machine learning is implemented for feature extraction and pattern recognition that can be used to develop intelligent systems. Machine learning is becoming more popular in healthcare as wearable devices and technology such as fitness trackers and smartwatches are progressively adopted by patients. These gadgets monitor customer health details and provide real-time medical evaluations to healthcare practitioners [15]. The recent advances in ML are utilized in deep learning. Deep learning is useful in studying neural networks. Complex sensory high-dimensional datasets of the brain are difficult to analyze. Thus, ML techniques can reconstruct these datasets to be real [14].

21.3.4 Reinforcement Learning

Reinforcement learning is a part of machine learning that is developed through neurosciencebased principles that train algorithms to learn from their environments. It is the science of decision-making, where an intelligent agent learns the optimal behavior in an environment to obtain maximum reward. Reinforcement learning is inspired by phenomena in animal learning, psychology, and neuroscience [19]. It has inspired computer programmers to develop novel algorithms for reinforcement learning in artificial systems to build applications like autonomous vehicles and robot-based surgery gaming stations. Reinforcement machine learning teaches computers to choose the best action through trial and error using a reward system. Reinforcement learning can educate models on how to play video games or teach self-driving automobiles to drive by informing the computer when it has made the right decision. The model learns through actions performed over time.

21.3.5 Neuron Networks

The neuron networks learn incrementally by supervisory feedback [29]. The comprehensive capabilities of the brain offer an attractive model for the design of AI or intelligent robots. Research on AI is mainly focused on the structural, functional, and mechanistic properties of the brain. Attempts have been made by mathematicians, scientists, and philosophers to model the information-processing mechanisms of neurons since 1950s. A highly reductionist model of neuronal signalling has been developed by researchers which is called perceptron. Perceptron is a multidimensional input-based biological neuron model utilized in artificial neural networks (ANNs). ANN includes interconnected nodes or neurons organized into layers. Neuroscience lays the foundation for the design of ANNs, which consist of nodes structured in input, hidden, and output layers. A neural network in its simplest form consists of an input layer, a hidden layer, and an output layer (the classification or prediction). The multiple nodes known as perceptrons of these layers are interconnected, and the number of hidden nodes will determine the complexity of the network as well as its risk of over-fitting. Different nodes of the network receive weighted inputs when data flows. These inputs produce a binary output when their summation reaches a specific threshold value. Therefore, predictions can be made according to input data or conditions [1]. However, the limitations of perceptrons have been rectified by multilayer neural networks and using backpropagation. Recently, neural networks have given rise to new-generation AI that is capable of recognizing speech and image. The backpropagation algorithm is normally used to train deep neural networks and tune the weights in a network to minimize errors [1, 7].

Backpropagation if combined with reinforcement learning methods, advanced learning systems such as strategy-based games can be created. Several biological replacements for backpropagation have also been suggested by authors such as feedback alignment, predictive coding, zero divergence inference learning, equilibrium propagation, and Hebbian-like learning rules [1].

21.4 History of Developments in AI and Neuroscience

AI and neuroscience fields are mutually interrelated. The long-term goal associated with AI is to create a human-level intelligent system. However, even after nearly 60 years of advancement in this area, an intelligent system capable of matching human intelligence has never been created. The significance of an intelligent system here means a system that can work with multiple distinct cognitive abilities, possess a significant capacity for adapting to complex surroundings, and possess the capability of self-learning to novel environments and new subjects [13]. Relevant data can be obtained via a small number of brain neural network (BNN) operations across a range of cognitive tasks at different scales [13]. Consequently, the field of computational science and AI has seen a rise in interest in the development of brain-like intelligence in recent years. Inspired by cognitive behavior mechanisms and achieved through hardware and software cooperation, the machine intelligence brain-like intelligence system (MIBS) works like the human brain [13].

21.5 Essential Technologies for Developing IAS

Earlier close-approach technology was utilized for the purpose of understanding and learning neurology. This technology is facing multiple difficulties as it has proven to be ineffective in determining psychiatric disorders.

The machine insights were found back in Turing, indeed prior to Pascal and Leibniz. In the embryonic period, Turing put forward a tall request for the advancement of AI systems that can think like humans. In 1956, some scholars put forward “Simulation, Extension, Expansion of Human Intelligence” and “Basic Definition and Long-term Objectives of “Science and Engineering for Manufacturing Intelligent Machines” [13]. After 60 years of research, the advancement that was made was the creation of theoretical principles in the practice of intelligent machines, such as the concepts and techniques of machine perception and pattern recognition. It also involves the creation of reasoning, representation of knowledge theory systems, theoretical and development of machine learning algo-

rithms, etc. [13]. In the "Dangerous Edge" Challenge, the IBM Watson Question Answering System (QA) outperforms human competitors like Siri. Other examples of advancements in this subject include the emergence of automated man-machine interaction and service systems, as well as Google Auto Driving [13]. Research initiatives are underway to create intelligent systems that resemble brains, such as those funded by the European Union. The Brain Program of the United States (BRAIN), also known as Project MICrONS (Machine Intelligence from Cortical Networks), was introduced in 2013 and is carried out by supercomputing [13].

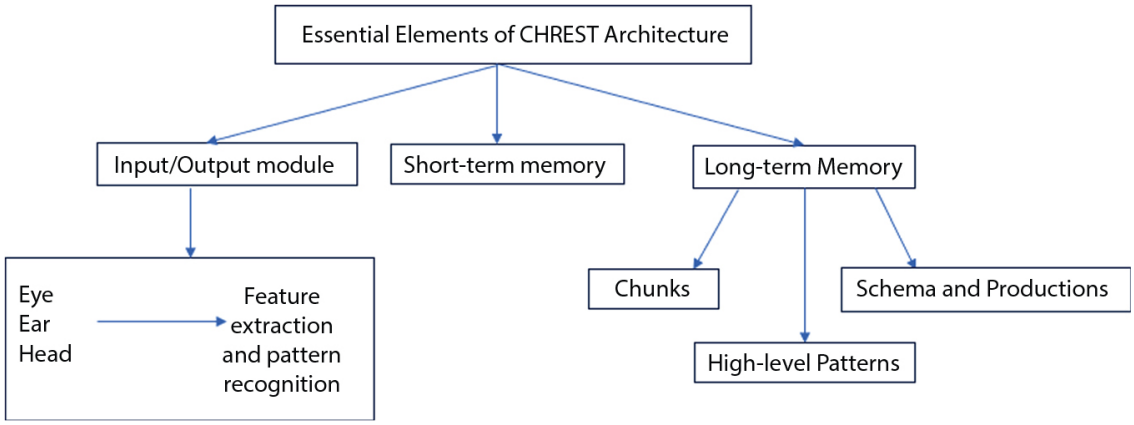


Figure 21.2 Essential elements of CHREST architecture.

CHREST architecture was used to develop artificial general intelligence. Its architecture presents a strong relationship between perception, learning, and memory. CHREST architecture consists of three main components. The first is the system and frameworks for communicating with the outside world. Second is the multiple STM (short-term memory) that stores data from a variety of input methods. The third component is LTM (long-term memory) which stores information in a "chunking network."

The essential elements of CHREST architecture are the input-output module and the short-term and long-term memory, as shown in Figure 21.2. All these elements are interrelated bidirectionally for interaction with the external world. The input-output module is useful for extracting features and recognizing different patterns in a similar manner as human being uses eye, head, and ear. Long-term memory retains visual patterns as well as high-level concepts learnt from input and output module experi-

ences/mechanisms and productions. However, short-term memory retains data temporarily.

21.6 Discoveries Related to Neuroscience

Various researchers have discovered systems and technologies for creating the largest resolution of 3D images of the human brain so that it could show vivid details of each cell and neural network connections in the human cortex. A few of them are listed below:

1. Cajal examined the polar configuration and structure of neurons as well as the cellular foundation of the nervous system.
2. Sherrington has described spinal cord reflexes and nerve circuits. Adrian's finding reveals that the frequency of action potentials encodes the amount of information in the brain.
3. Hodgkin and Huxley exposed the ionic component of activity possibilities and different neurotransmitters. They too have made the primary fine activity potential demonstrations on physiological tests.
4. Katz and Eccles performed the analysis of chemical synaptic transmission. Hubel and Wiesel studied the characteristics of different visual neurons from straightforward to complex open areas.
5. Bliss and Ito disclosed long-term synaptic upgrades and debilitating, O'keefe's revelation of neurons responsive to particular spatial localization [13].

These disclosures led to demonstrate the reality that brain science studies brain architecture and individual neuron operation. However, cognitive science is the scientific and interdisciplinary study of the mind and its processes. The neuronal activity in the brain can be increased artificially by indenting light on genetically altered neurons. Moreover, the convolutional neural arrangement comes from the visual cortex of a cat's brain. Despite significant progress in neuroscience research, understanding of brain disorders is still limited.

21.7 Applications of Artificial Intelligence in Neuroscience

AI has offered significant potential in neurology transformation as diagnoses and treatment procedures of brain diseases can be enhanced in terms of analyzing large volumes of data. Algorithms based on AI are harnessed to enhance the accuracy and neurological diagnosis speed [32]. The primary step is the high-level unique depiction of human thought and behavior. Research on artificial neural networks began in the 1940s. This was the initial model of the brain's nervous system's structure and computer mechanism in which connectionism was represented by an artificial neural network [13]. The first application of AI technology in the domain of neurosciences is that it helps in studying the processing of a person's brain. The second application of AI technology is that it explains the secrets of the human brain and how it functions. The third application of AI in neuroscience study is to provide information about neural patterns that make the study of neuron structure easy for neuroscientists. All these applications provide an analogy of the brain working. Also, machine learning recognizes the data provided by the inquiry of neural patterns. The complex structure of the working of the brain is studied by analyzing the brain using machine learning technology. The mystery of how millions of nerves and neurons work is solved using machine learning as well as AI technologies for studying neurology. Machine learning technology, known as imaging and resonance technology, provides information about neuronal and brain activity. The flow of blood in the brain is also studied using imaging and resonance technology. The main working mechanism of imaging technology is that it provides a three-dimensional and high-quality image of neurons that indicates their structure and function in brain activity. The 3D information about neurons then helps in speeding up the brain or neuron-related research work. Treating a disorder through AI technology improves the patient's condition and increases the chances of speedy recovery of the patient. Researchers explained that using technologybased systems in the medical field increases job opportunities for professional physicians or neurologists. Only a well-trained neurologist or physician can work with AI-based devices for treating neuron-related diseases in patients. Studies show that to improve services in the health sector, technology-based integrated AI systems provide immense

applications. From monitoring the health condition of patients to diagnosing the patient's health condition are the duties of the AI-based health sector.

Computational deep learning intelligence algorithms used in clinical processes allow physicians to identify and treat the tumor at an early stage. Researchers show that the size of a brain tumor varies in its stages. The stage of a brain tumor is identified by predicting the tumor size with the help of various deep learning and machine learning algorithms in AI technology systems. Some radio imaging technologies also predict the size of tumor formation in the brain for identifying the tumor age. Researchers studies explained that in the neurosurgery process, the equipment used works on the principle of AI technology for providing better surgery performance. Improved surgery performance then results in faster recovery of the patient after surgery. By predicting the type of brain tumor through intelligence-based techniques, it becomes easy for the neurologist to perform surgery and then prescribe high-precision medicines. Scholar's studies revealed that for theorizing the neurological data, deep learning, and neural network-based methods are merged with AI.

The merging of these methods with AI provides authentic data and easy-to-understand information regarding the site of neuronal damage. researchers explained the scope of AI in the medical sector by considering it to be the technology used for enhancing medical services. The work of scholars depicts that all the therapeutic processes for treating neurological disorders rely on the extent of the use of AI in neurological health sectors [21]. Most neurosurgeons refer to the use of AI-based chemotherapy techniques for treating tumors. Researchers claiming the efficiency of AI technology in diagnostic processes predicted that AI-based machine learning systems are the most efficient systems used for the recognition of neurological disorders [15].

21.7.1 Creation of Chip-Based Intelligence Similar to the Brain

Chips that resemble brains and computing hardware systems are being developed to create intelligence comparable to the brain and to achieve high-performance low-power computing systems. Its operating concepts

are based on the brain-nervous system and utilize the information-processing method of neurons.

21.7.2 Projects Based on the Development of Brain-Like Intelligence

The decade of 1990–2000 is referred to as the "Decade of Brain" due to the fact that the United States started a program named "Brain Research Program" in the 1990s. The Blue Brain Project was started by Switzerland's Federal Polytechnic University of Lausanne (EPFL) [13].

1. The SpiNNaker project is a brain-like intelligence project at the hardware level that used an ARP chip and learnt from neuron discharge mode.
2. In 2014, the IBM Company created a chip named TrueNorth chip that can integrate memory and computation by gaining knowledge from the functioning of neurons. This chip includes 406 nuclei, 1 million neurons, and 2.56 neurons.
3. A neural network-based chip that uses only memristors was created by Prezioso *et al.* from the University of California and Stony Brook, New York State University.
4. A project named Brain Scale S project was suggested by Germany's Heidelberg University which integrates super-large-scale synapses on crystals to enhance computational power and lower communication expenses. High-performance computing and brain-like mechanisms have advanced because of this project.

DaDianNao developed deep learning-based chips that attained 450.65 times faster than a single GPU chip in the field of deep learning and 150.31 times less power usage for 64 nodes. To enhance the current deep learning chips, it seems likely that brain informationprocessing mechanisms will be incorporated in the future [13].

21.8 Artificial Neural Network Versus Biological Neural Network

Neural networks have been developed representing biological processes [27] analogous to evolutionary algorithms. The neural connections in the human brain were revealed with the use of a microscope. Thus, the ANN model was developed in an AI domain. Moreover, the multilayer structure and convolution properties of the brain were discovered using electronic sensors/detectors inspired by deep learning and convolutional neural networks [7].

The neural network is made up of an input layer (data features), a hidden layer, and an output layer (prediction and classification). Several nodes of these layers are interconnected; complexity of the network is determined by hidden nodes [10]. Figure 21.3 describes a basic artificial neural network which is also called perceptron. Here, x denotes the binary inputs. w denotes weights. The layer between is the hidden layer that transforms inputs into single binary outputs labeled as y. The two central blocks present in between the weights and output y describe the rules for transforming the input into a single binary output. At first, each input is multiplied by its weight, and then all multiplied terms are added. Then, a bias (constant) is added.

Before the development of deep learning, there was an algorithm called the backpropagation algorithm that showed excellent functionality in non-linear pattern classification that caused a spike in research and applications of neural networks. When GPU parallel computing was introduced, more training was carried out on vast amounts of data, and layer neural networking was highly suitable for training and generalization of neural networks. However, an artificial neural network with various layers is not efficient for learning abilities [13]. The CLS (complementary learning systems) theory is a theory that highlights links between the principles of CLS theory of human learning and memory which has proven that it is a very useful theme in machine learning, comprising neural network architectures that incorporate memory modules [18].

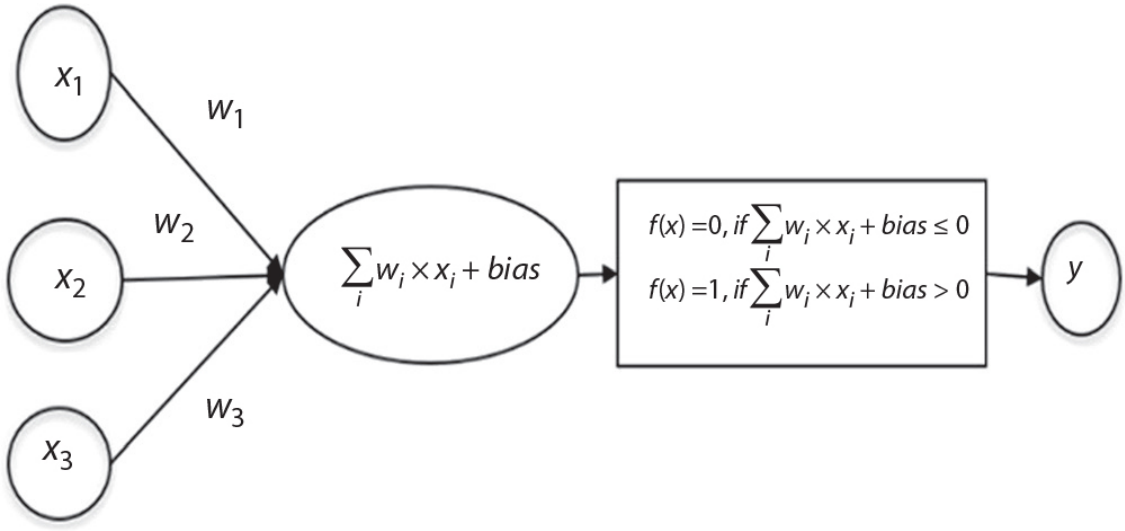


Figure 21.3 Artificial neural network

(reproduced with permission from authors) [14].

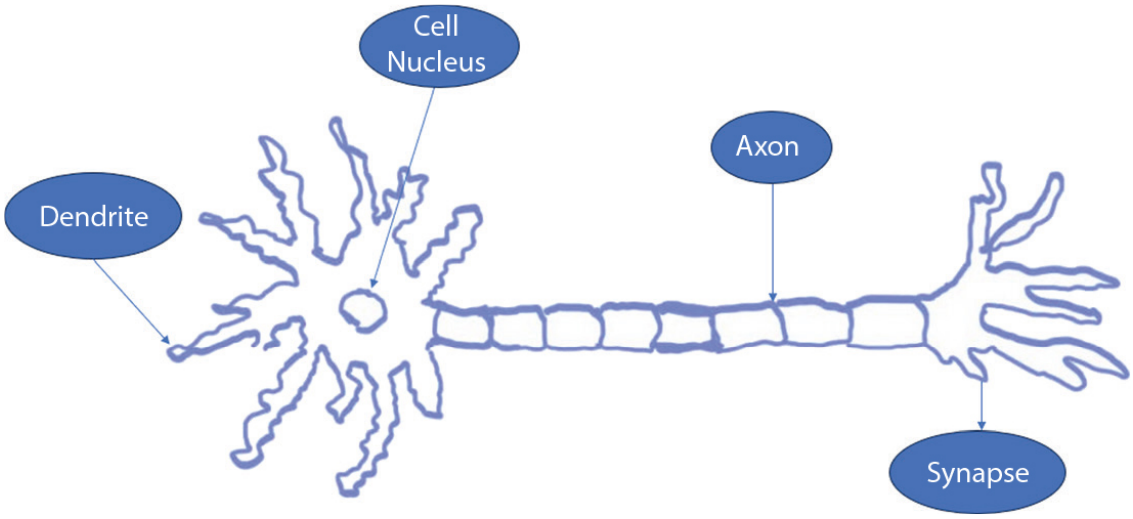


Figure 21.4 Biological neural network.

Within our biological brain, visual information gets converted from pixel inputs as activation of photoreceptors in the retina to neural responses in subsequent visual cortical areas along the visual pathway. The chemicals present in the neurons do the processing. The biological network does massive parallel operations [20].

Similarly, ANNs as the name describes, are the artificial neural networks that are obtained from biological neural connections of the brain. ANNs consist of neurons that are connected to one another in different layers of the networks, just like neurons do in the human brain. These neurons are known as nodes. The control unit does the processing [14]. These artificial neural networks and biological neural networks are both capable of enormously parallel computations.

In Figure 21.4, dendrites represent inputs, the cell nucleus symbolizes nodes, the synapse symbolizes weights, and the axon represents output in a biological neural network. The invention of the microscope helped in understanding the structure of the brain and interconnected connections among neurons. However, AI and machine learning can produce large-scale simulations of neural processes that generate intelligence. The biological neural network has led to the realization of complex deep neural network architectures that can be used to develop versatile applications, such as text processing, speech recognition, and object detection.

21.9 Developments of Intelligent Automation Systems Models

Various models and architectures have been used in robotics and autonomous systems. Poggio *et al.* worked on the HMAX model which is a biologically inspired architecture that imitates human visual information processing. This model is capable of achieving numerous rapid object recognition tasks or computer vision similar to human recognition performance.

21.9.1 LIDA (Learning Intelligent Distribution Agent) Architecture

Intelligent automation is the primary application of neuroscience. An architecture named LIDA architecture is a software system that is employed for the personnel work for the US Navy. The LIDA architecture is an extended version of the IDA (Intelligent Distribution Agent) model. LIDA, the learning IDA, will add three types of training to IDA: perceptual training, episodic training, and procedural training. The model enables a fine-grained analysis of a wide variety of aspects of human cognition, organized as a cycle of processes, beginning with perception, and ending in an action. The LIDA computational model generates a hypothesis at the mechanism level. These hypotheses are typically in the areas of functional designs and their underlying requirements, independence, interdependence, and communication among modules that correspond to counterparts in neurobiology [17].

21.9.2 Human-Machine Interaction (HMI)

Human-machine interaction is defined as the exchange of information and communication via a variety of interfaces between human users and machines in a dynamic context. Businesses analyze real-time data using a variety of algorithms. While fewer people will be needed to handle big data, an industrial data scientist proficient in statistical programming languages will still be needed. Businesses gain a competitive edge when they are able to manage large amounts of data. Robots' cognitive ability to adjust their status in response to changing circumstances and novel inputs will play a crucial role in the dynamic environment of the production process [23].

21.9.3 Cognitive Automation

Cognitive automation refers to seizing ML to automate cognitive knowledge and service work to realize the value offered by AI. This advanced form of automation is based on implementing artificial cognition that mimics and approximates human cognition in machines [24].

Nowadays, AI is becoming a vital tool in neuroscience research for performing intelligence-based actions [1]. The fundamental hypothesis in AI can be stated that computational entities of cognitive systems are minds. Cognitive systems are able to perceive, understand, draw conclusions, and learn things just like human capabilities and skills. These systems can be implemented using a variety of physical devices [2]. Cognitive science, AI, robotics, and neuroscience contribute to the understanding of minds. Cognitive science involves the modeling of natural minds, thus concerns for the understanding of cognitive processes lead to human thought. The standard model reproduces real consensus over the assumptions and provides a platform in a complete cognitive architecture by integrating theoretical ideas without realization.

Figure 21.5 shows the core components of the standard model. The illustration of its different blocks is mentioned below:

1. Declarative memory: It is a long-term memory that stores facts and can be a repository of direct experiences in the form of knowledge.

2. Working memory: It provides temporary global space for retaining the outputs of long-term memory and perception. Additionally, it includes buffer space for storing the output of declarative memory and motor actions. In brain-like AI systems, a major obstacle is to model working memory as an intelligence component [1]. Working memory was discovered from the results obtained by functional Magnetic Resonance Imaging (fMRI) that encourages memory modules in the machine learning module.
3. Procedural memory: It is based on pattern-directed supplication of actions and contains information about internal and external actions. It induces specific processing that is required to select a single cautious act per cognitive cycle.
4. Perception: It converts external input signals into relations and symbols and places the results within working memory. Inputs can be visual or auditory signals.
5. Motor: It converts symbols that are stored in buffers into output actions that are part of the biological body, like the movement of arms and legs.

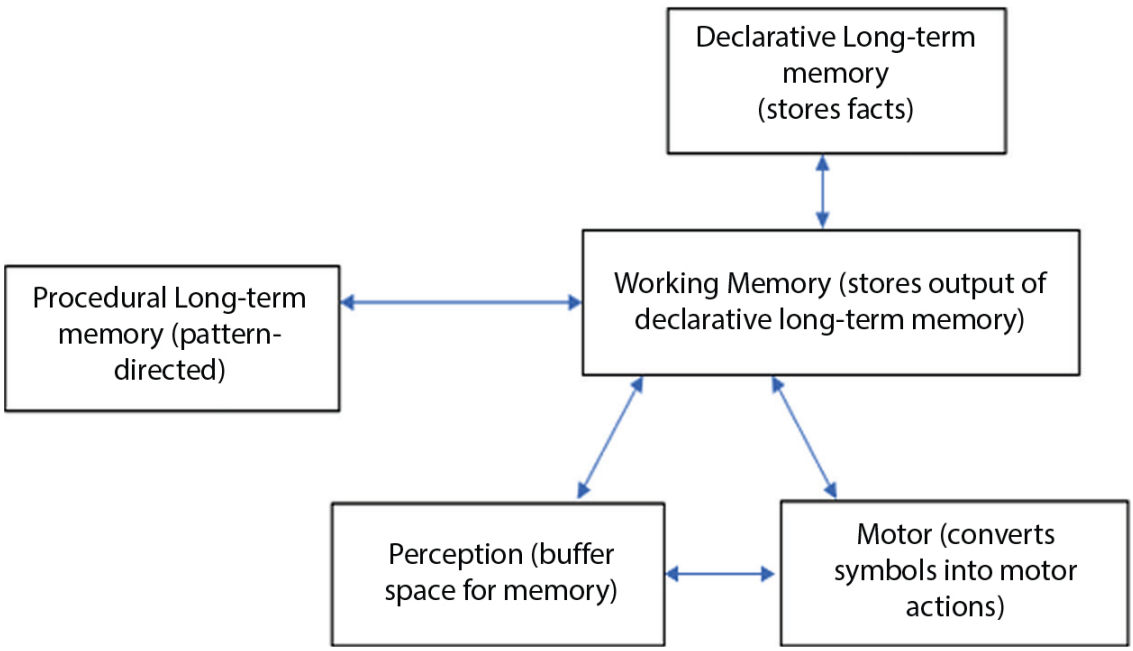


Figure 21.5 Structure of standard model.

The heart of the standard model is a cognitive cycle that starts from collecting input, processing, storing, and providing output. Sequences of this type of cycle are used to analyze external and internal complex behavior [2]. AI concerns about building artificial minds could exhibit intelligent

behavior. Robotics relies on designing and controlling artificial bodies through minds. Neuroscience studies the structure and functions of the brain and cares for the arise of minds from brains.

Advances in machine and deep learning algorithms are now used to recognize unusual patterns of brain action and to simulate dysfunctional and normal states of the brain. Novel AI systems led to the development of high-dimensional deep neural networks that help to foster advancements in the neuroscience field [1]. The fundamental problems associated with neuroscience can be explored using high-throughput data analysis methods [7].

21.9.4 Emotion Recognition

Emotions are essential to human nature and have a significant impact on thought processes. Emotion is frequently linked to perception, human interaction, reasoning, and, to some extent, human intellect itself.

This subject addresses the components of research size, EEG (Electro Encephalo Graphy) devices, machine learning classifiers, exhibition strategy, emotion stimuli kind, and classification technique [25]. Emotion recognition software uses AI and ML algorithms for analyzing and interpreting facial expressions as an outcome of emotions.

21.9.5 Continuous Learning

Intelligent systems are required to remember and learn different functions that are coming across at several timescales. Thus, artificial and biological agents must have continuous learning capacity used for continual learning. Continual learning is the ability to master novel functions without forgetting prior tasks. Neural network parameters shift in the direction of the optimal state for performing consecutive tasks but overwriting the first completed configuration. Therefore, these networks suffer from catastrophic problems [29].

Brain-inspired intelligent robotic systems not only contain musculoskeletal structures but also have visual recognition, decision-making, and mo-

tor or action control abilities [4]. Brain inspired processing algorithms based on intelligence and robots with high computational efficiency can be designed by integrating AI, robotics, and brain science disciplines. Robots can be equipped to adapt to new environments through continuous learning ability or motion-learning algorithms. Therefore, brain-inspired control and cognitive algorithms aid in next-generation robots [4].

21.10 AI for Neuroscience Development

The key strength of AI is to analyze a bulk amount of data and to extract hidden patterns. If a known algorithm is discovered to be implemented in the brain, neuroscience would be a strong support for its viability as a component in large general intelligence systems. Neuroscience can be used to improve the quality of algorithms in AI. In order to generate results compared to a human brain, data experts need to create benchmarks regarding the effectiveness of AI algorithms.

Thus, AI assists as the most suitable choice for extracting patterns and interferences of complex brain signals by producing simulations of neural processes [14].

21.10.1 Brain-Computer Interfaces

Brain-computer interface (BCI) is one such emerging technology in Neuroscience. In a nutshell, BCI technology provides direct communication between the brain and external devices bypassing the normal neuromuscular pathways. BCI research and neuroscience have enabled the development of a precise understanding of decision-making [8]. Human-automation interaction field infers the integration of automated tasks into a self-governing system [9]. Automated systems provide time-sensitive and safety settings so that issues related to task and function allocation between machines and humans, and trust issues can be resolved.

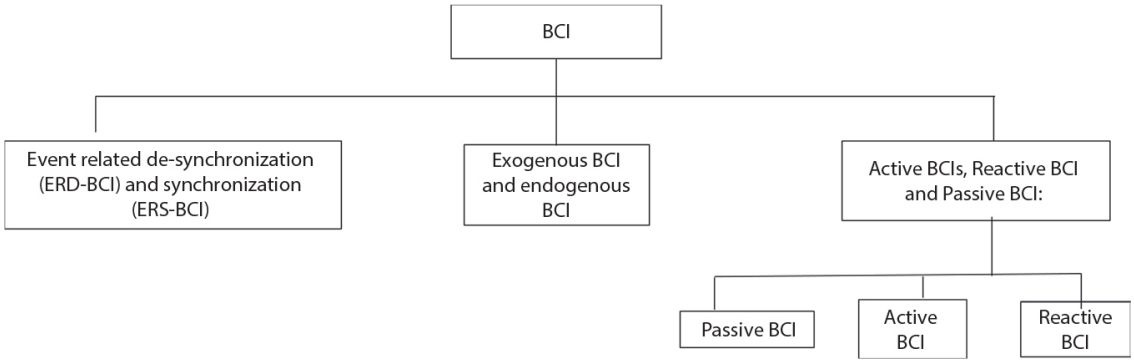


Figure 21.6 BCI and its types.

BCI not only serves the medical field & health care but also has a role in various other arenas of human life like entertainment, gaming, education, self-control, and marketing. Brain-computer interface with AI is a rapidly growing new technology that provides a direct means of communication between a silently speaking brain and bio-monitoring devices, smartphones, earphones, and other external devices.

A fast-emerging new technology called brain-computer interface using AI allows a silently thinking brain to interact directly with smartphones, earbuds, biomonitoring devices, and other external devices. The brain working together with a device to provide straight electrical signal transmission from neurons to an external equipment or system, like a computer or robotic arm, is known as a brain-computer interface. It is a potent communication technology that carries out the order and, consequently, the action without requiring the use of any muscles or neuromuscular circuits. Figure 21.6 shows the major categories and sub-categories of a BCI system.

The components of BCI are signal acquisition, feature extraction, translation, and device output.

1. **Signal Acquisition**: The initial component of the BCI system, signal acquisition, senses and measures brain impulses. The primary function of this component is to receive and record the signals produced by neural activity. It then transfers these signals to the BCI's next component, the preprocessing element, to enhance the signals and attenuate electrical noise. Three popular techniques are employed for acquiring signals.

1. **Invasive:** This kind of BCI involves implanting tiny electrodes directly into the cortex of the brain to measure neural activity.
2. **Semi-Invasive:** Electrodes are placed in the arachnoids' space or the epidural space in this kind of BCI.
3. **Non-Invasive:** This kind of BCI involves applying electrodes to the scalp. EEG is the most widely used non-invasive technique; although, more recently, MEG (magneto encephalo graphy), PET (positron emission tomography), fMRI, and fNIRS (functional near-infrared spectroscopy) have been used [34].

2. **Signal Processing:** The initial stage of the BCI's signal processing is feature extraction. This task involves analyzing digitally generated signals to separate relevant signal characters from unnecessary data and presenting signals in a way that makes sense for output commands. By limiting the amount of data that may be used for the subsequent component, this component creates the enhanced signal's selectable features.
3. **Device Output:** The output device is directed and operated by the instructions obtained from the translation process. It assists with achieving the user's goals, such as choosing an alphabet, controlling a mouse, moving a robotic arm, operating a wheelchair, moving a paralyzed limb with a neuroprosthesis, and so on. Currently, the most implemented output device for communication is a computer. Figure 21.7 describes BCI and its components and its working starting from signal acquisition processing, controlling the output, and then feedback to input for minimizing any error. Control signal is taken by a control system or mechanism and feedback is provided to the biological mind.

21.10.2 Neuroimaging Analysis

For a more comprehensive study of the brain and its development investigation, noninvasive neuroimaging technologies are largely contributing [5, 6]. Machine learning models help in analyzing medical imagery data such as MRI scans and detecting anomalies and disorders with extraordinary precision [32]. Deep neural networks are used for performing different upstream tasks like resolution improvement, noise removal, image normalization, and image reconstruction [14]. The structural development of the brain involves changes in thickness, volume, and synaptic

density of gray and white matter. The structural and functional images of the brain can be visualized using MRI (magnetic resonance imaging) techniques that provide information about size, shape, fiber orientation, size of white matter, etc. The facial or visual recognition, blood oxygenation level, and functions associated with social cognition are related to the functional development of the brain [5].

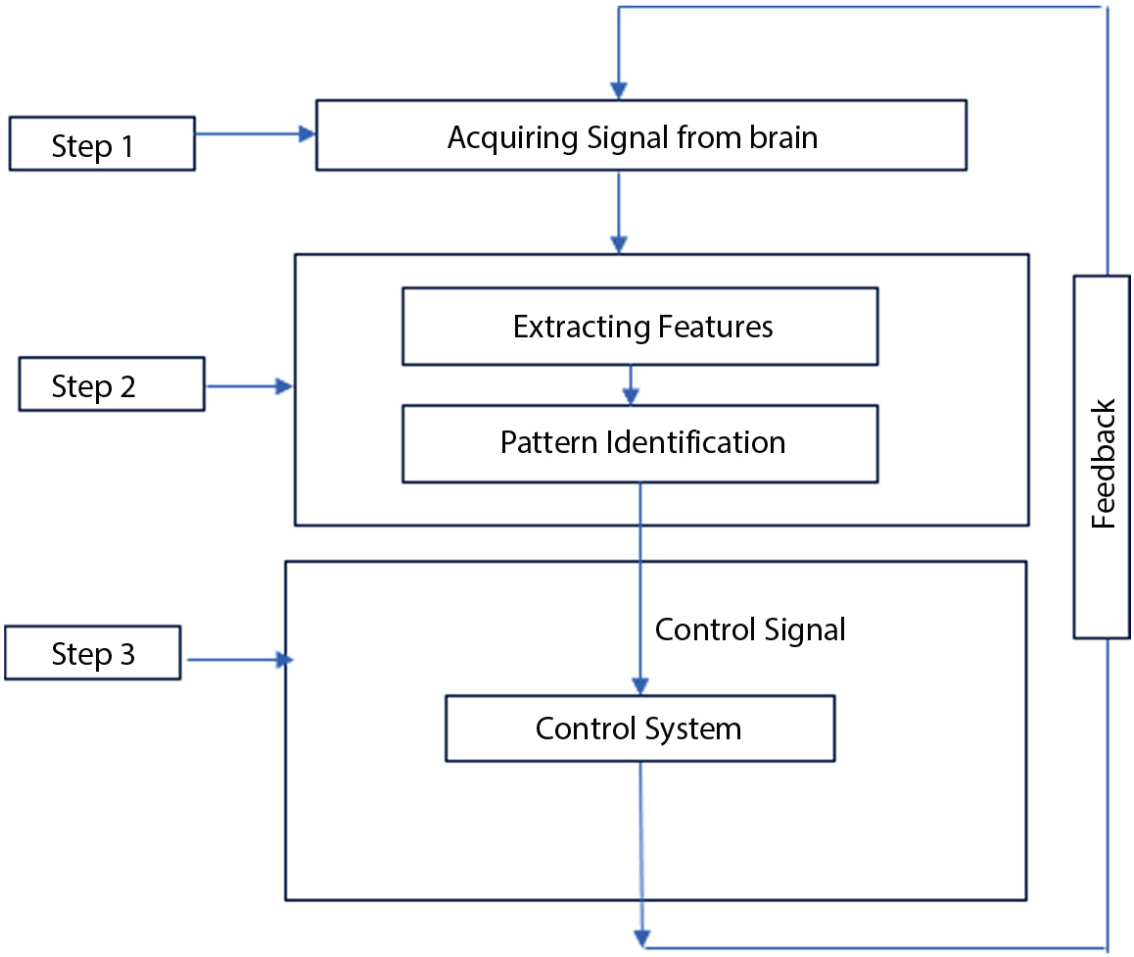


Figure 21.7 Brain-computer interface model.

21.11 Neuromarketing

Neuromarketing, a useful application of the non-invasive BCI innovation, has developed as an intriguing bridge between neuroscience and marketing that has transformed the idea of marketing research. The medium that connects a product to customers and ultimately determines the sale is called marketing. A good product cannot inform, engage, and maintain its intended audience in the absence of efficient marketing. To thrive in a competitive marketplace, recently released products require even more intense marketing.

Classical marketing, on the other hand, only provides insights into customer behavior after the fact. In order to gather consumer input, traditional market research employs surveys, focus groups, in-person interviews, field tests, and observations. These methods have the drawbacks of requiring a lot of time, being expensive, and using shaky data that frequently yields incorrect answers. Unlike conventional marketing research methods, neuromarketing forecasts customer purchase decisions and captures consumers' implicit cognitive and emotional responses to a range of marketing stimuli.

Neuromarketing has replaced traditional survey methods by directly measuring the brain response of a client to marketing stimuli using non-invasive brain signal recording techniques. The brain's neural recording instruments utilized in neuromarketing research consist of EEG, fMRI, transcranial magnetic stimulator (TMS), MEG, PET, fNIRS, etc. These gadgets enable the exploration of a consumer's cognitive and psychological reactions, such as like/dislike and approach/withdrawal, by obtaining neural activity from the human brain.

The brain of humans reacts to various stimuli in different ways, and the response can be observed by monitoring changes in neural activity of the brain or neuronal impulses. Furthermore, the researchers have been able to monitor, examine, and interpret the potential interpretations of brainwaves due to advances in machine learning algorithms and signal and image processing techniques. This introduces new opportunities and possibilities for researchers in marketing to identify, evaluate, and forecast consumer purchasing patterns [28]. The client's emotional states such as enthusiasm, engagement, withdrawal, and stress while they are exposed to a marketing stimulus can now be recorded with the help of brain-computer interface technology. Eye monitoring, heart rate monitoring, galvanic skin response, facial activity coding, and other methods, with or without brain signal recording methodologies, have been employed in several neuromarketing studies to measure the consumer's concealed response. An item's reaction to the consumer can be identified by their direct interaction with it or by seeing a photograph of it.

Promotion is the advertising used by marketers to manipulate consumers' decisions about what to buy. Promotion is typically identified in neuromarketing research as TV ads and short films. Assessing the level of customer involvement with commercials is one of the main priorities for neuromarketing. Estimating the level of engagement with advertising prior to their airing guarantees a higher success percentage for promotional campaigns. The majority of the TVCs (television commercials) in these books had a uniform duration of thirty seconds. These TVCs were linked with other videos documentaries, gaming videos, dramas, etc.

In neuromarketing, in order to gauge the actual reaction of viewers, research on neuromarketing, since its operation, is closely linked to the interpretation of neurological response. The brain uses action potentials, or the firing of neurons, as a means of neural communication. The electrical information that neurons transmit to one another is called a neural signal. These data are obtained as brainwaves, which are non-linear pattern signals. The neuronal signature of various mental states is also connected to these brainwaves. The frequency bands that make up the neural signature are called rhythms; these include the delta (0.1–4 Hz), theta (4–8 Hz), alpha (8–12 Hz), beta (12–30 Hz), and gamma (30–90 Hz). Beta wave amplitudes are linked to reward processing in neuromarketing, which helps forecast a product's success even more.

Preprocessing neural signals to improve the signal-to-noise ratio (SNR) is essential before beginning any analysis or interpretation since neural signals and images are extremely susceptible to noise and artifacts. Heart rate variability using ECG, power line interference, eye movement artifacts using electroocoulogram (EOG), and muscle movement artifacts using electromyography (EMG) are among the noises that frequently accompany EEG signals.

In neuromarketing, preprocessing includes screening the signals to the desired frequency bands, re-referencing the signal that was filtered to a shared average, identifying and interpolating problematic channels, removing noise and artifacts, and framing or segmenting the data in preparation for additional machine learning steps.

Neuromarketing utilizes both supervised and unsupervised machine learning. In supervised learning, test subjects' like/dislike reactions during interviews serve as the a priori ground truth in neuromarketing. In the training datasets, the labels aid the classifier in understanding the signal pattern of like and dislike EEGs. In the testing stage, a dataset without labels is used for predicting likes and dislikes. The training dataset labels can be hidden from the classifier by the scientist and then used to calculate accuracy. On the other hand, prior knowledge of the like/dislike labels is not necessary for the unsupervised learning approach utilized in neuromarketing. The signals are analyzed with the goal of inferring the current structures for various classes. Supervised learning typically resolves regression or classification issues.

The four most popular supervised learning classifiers in neuromarketing are support vector machines (SVM), random forests (RF), naive Bayes, and ANNs. Simultaneous unsupervised learning in Neuromarketing has made significant use of clustering-type classifiers, including independent component analysis (ICA), principal component analysis, K-NN (k-nearest neighbors), and singular value decomposition [26].

21.12 AI Inspired by Brain Science

The objective of AI is based on investigating the theories and developing computer systems that can conduct biological or human intelligence tasks or functions such as recognition, perceptrons, decision-making, and action control. On the other hand, the goal of brain science or neuroscience is to study the structure, functions, and operating mechanisms of biological brains [1, 7]. AI is considered as the simulation tool of brain intelligence if combined with Psychology and cognition science. Microscopy imaging instruments can establish a link between the structure, macrocognitive decision, and function of neural networks. Therefore, human intelligence and AI can be related. In addition, a new facet of biological computations is recognized to be critical in support of cognitive function. This is considered an excellent to be incorporated in artificial systems [29]. A known algorithm utilized within the brain strongly validates its feasibility as an essential element of a comprehensive general intelligence system.

The BRAIN initiative launched in the US aims to map dynamic brain images that exhibit complex and rapid interactions between surrounding nerve organs and brain cells. On the basis of multidimensional relationships, the processes of applying, processing, recording, storing, and recovering information in the brain can be understood.

Neuroergonomics is a multidisciplinary field that studies the relationship between human brain functions and performance or behavioral outcomes at work in everyday life [6]. In other words, high workload mental pressure results in impaired performance due to limited neurological mechanisms that prioritize a precise goal. Basically, the complexity of brain dynamics and networks can introduce characteristic limitations on the processing of information by the human brain. Therefore, there is a need to target neurophysiological states and associated mechanisms of the brain that are accountable for degraded performance. Authors in this research work have proposed various solutions to initiate a change in human behaviors by adapting user interface, task, level of automation, and neuro-adaptation [6].

21.13 Current State

Machine learning innovation including AI has great applications in the medicinal domain as they provide potential applications for the advancement of the health field. Currently, intelligence-based computational systems are used in healthcare monitoring systems. These devices are recently being utilized to study neurology. Moreover, research in neuroscience is done using these intelligent systems as well. These intelligent systems are built using AI and machine learning. These are employed for the purpose of understanding neurotransmitters in mental health. BMI (Brain Machine Interface) technology is now utilized for transmitting information between external devices and the brain.

Neuroscience helps in developing AI. Neurologists identify the disorder or neuron injury with the help of various machine learning and AI-based methods. It even helps in detecting the reason for damage. This highly efficient AI technology in neurological study allows the high-resolution

study of the brain for identifying the change in brain functionality of a patient affected by neuronal damage.

The high efficiency of AI technology in neurological study allows the high-resolution study of the brain for identifying the change in brain functionality of a patient affected by neuronal damage. Computer-based imaging techniques help in examining the overall structure of neurons to make neuron-related studies easy for neurologists or neurosurgeons. Moreover, neurosurgeons can detect damaged neurons only by using machine learning-based or artificial-based medical devices in the surgery process. In the health sector, the research on neurons and nerves by neurologists is made easy through the use of AI-based technology algorithms and models. The research on neurons is conducted for testing and understanding neuron-related disorders in patients and for improving their brain functionality and overall neurological health. Many countries all over the globe are making concerted efforts to advance their neurological healthcare sector, advanced by introducing AI as well as machine learning techniques in health sectors. These technologies are being introduced to enhance diagnostic accuracy, predict disease progression, and personalize treatment plans. The global push toward incorporating AI and ML in neurological issues signifies a promising future where technology and medicine work hand-in-hand [15].

21.14 Conclusion

For understanding and enhancing human intelligence, the realms of AI and neuroscience have laid down a profound relationship. AI, ML, and neural networks serve as powerful tools for modeling and simulating cognitive processes. Current AI research provides deep learning and reinforcement learning platforms. Neuroscience benefits from AI and encourages designing systems emulating human intelligence. Complex neural networks help in tackling process revealing and computational challenges underlying perceptual skills. AI technologies have proved amazing potential in enhancing human cognition, memory, various aspects of learning, decision-making, and problem-solving. BCIs aid in establishing direct communication between the brain and external devices enhancing cognitive functions. Thus, recent advances in automation technologies

are moving toward replicating human intelligence. Novel algorithms and enhanced computational power could drive innovations in AI-brain interfaces. The continued collaboration between neuroscience and AI will certainly expedite the progress in both areas.

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Note

1. *Corresponding author: harpreet.kaur@chitkarauniversity.edu.in